

BACKGROUND & MOTIVATION



Large Language Models (LLMs) are increasingly used in high-stakes domains such as **finance**, **healthcare**, and **decision systems**.

Despite their fluency and confidence, LLMs often exhibit:

HALLUCINATION



Factually incorrect answers

INCONSISTENCY



Conflicting responses to the same query

EVASION



Avoiding or deflecting uncertain questions



These issues are difficult to detect because responses appear **highly convincing**.



THE PROBLEM

- ▶ Traditional evaluation methods focus primarily on **accuracy**.
- ▶ They fail to capture deeper **reliability** issues.
- ▶ In high-stakes domains, even a single confident incorrect response can lead to **serious consequences**.



THE MOTIVATION



There is a need for a system that goes **beyond accuracy**



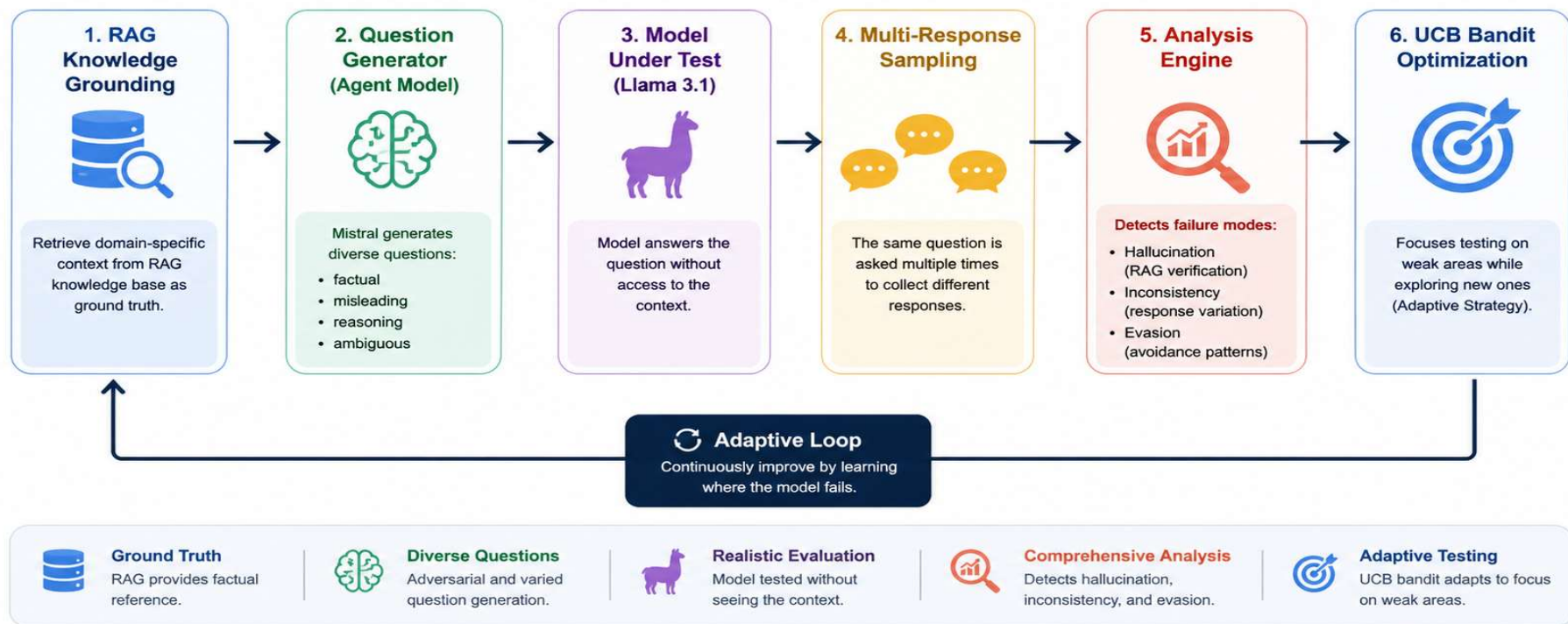
And evaluates the **trustworthiness** of LLM behavior

“ A **confident wrong** answer is more dangerous than no answer. ”

APPROACH

LLM Stress Tester – System Overview

Adaptive evaluation of Hallucination, Inconsistency & Evasion



DESIGN CHOICES & RATIONALE

We made specific design choices to ensure realistic, robust, and adaptive evaluation of LLM behavior.

01 RAG-based Ground Truth



RAG

- ✓ Retrieves relevant domain-specific context.
- ✓ Provides reliable ground truth for factual verification.

02 Separate Generator & Test Models



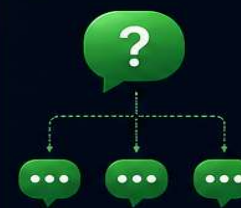
Mistral
(Generator)



Llama 3.1
(Test Model)

- ✓ Mistral generates diverse questions.
- ✓ Llama 3.1 is evaluated.
- ✓ Prevents bias and information leakage.

03 Multi-Response Sampling



- ✓ Same question is asked multiple times.
- ✓ Enables detection of inconsistency in responses.

04 Evasion Detection Logic



- ✓ Detects when the model avoids or deflects answering.
- ✓ Captures real-world failure patterns.

05 UCB Bandit Algorithm



- ✓ Focuses testing on weak areas.
- ✓ Balances exploration (new areas) and exploitation (weak areas).
- ✓ Improves efficiency of testing.

06 Why These Choices Matter



- ✓ Realistic evaluation without leakage.
- ✓ Detects multiple failure modes.
- ✓ Adaptive and data-driven stress testing.



Key Principle

The system evaluates behavior under **realistic conditions**, not just **correctness**.



Reliable Ground Truth



No Information Leakage



Detects Hidden Failures



Adaptive & Efficient

RESULTS

FINANCE DOMAIN

HALLUCINATION REPORT

Model : llama3.1:latest
Domain : finance
Steps : 30

Overall hallucination rate : 20.0%
95% confidence interval : [9.5% - 37.3%]

Failure Breakdown:

Hallucination : 2 (6.7%)
Inconsistency : 7 (23.3%)
Evasion : 4 (13.3%)

Category	Rate	CI (95%)	Samples	Verdict
factual	40.0%	[16.8% - 68.7%]	10	⚠ Moderate risk
misleading	0.0%	[0% - 39.0%]	6	✅ Reliable
reasoning	14.3%	[2.6% - 51.3%]	7	🟡 Low risk
ambiguous	14.3%	[2.6% - 51.3%]	7	🟡 Low risk

CONCLUSION

⚠ Use llama3.1:latest with caution.

Weakest area : factual (40.0%)

Full results → results.jsonl

EDUCATION DOMAIN

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

(llm_tester) (base) qayyumkhan@Qayyums-MacBook-Pro llm-stress-tester % python main.py

Wikipedia : ✓ - CORRECT - The process of converting pyruvate into acetyl-CoA is indeed a function
Hallucination: ✅ NO

FINAL SUMMARY

Total steps : 50
Hallucinations : 3 (6%)

Weakest areas (most hallucinations):

factual	21% (17 samples)
misleading	8% (11 samples)
reasoning	8% (11 samples)
ambiguous	8% (11 samples)

Full results saved to: results.jsonl

o (llm_tester) (base) qayyumkhan@Qayyums-MacBook-Pro llm-stress-tester % python main.py

Ln 36, Col 30 Spaces:

RESULTS

PHYSICS DOMAIN

```
=====
HALLUCINATION REPORT - PHYSICS DOMAIN
Model      : llama3
Domain     : physics
Steps      : 200
=====

Overall hallucination rate : 4.0%
95% confidence interval   : [2.0% - 7.7%]

Failure Breakdown:
  Hallucination : 1 (0.5%)
  Inconsistency : 16 (8.0%)
  Evasion       : 6 (3.0%)

Category      Rate  CI (95%)      Samples  Verdict
-----
factual       12.5% [6.2% - 23.6%]  56       🟡 Low risk
misleading    0.0% [0% - 9.9%]   35       ✅ Reliable
reasoning     0.0% [0% - 9.9%]   35       ✅ Reliable
ambiguous     2.6% [0.5% - 13.2%]  39       ✅ Reliable
numerical     0.0% [0% - 9.9%]   35       ✅ Reliable

CONCLUSION
✅ Reliable for physics content.

Weakest area : factual (12.5%)

UCB1 BANDIT BEHAVIOR
Category      Selected  Halluc%  % of Total  Bandit Signal
-----
factual       56        12.5%    28.0%      ⬆ Focused
misleading    35         0.0%    17.5%      ⬇ Reduced
reasoning     35         0.0%    17.5%      ⬇ Reduced
ambiguous     39         2.6%    19.5%      ⬇ Reduced
numerical     35         0.0%    17.5%      ⬇ Reduced

REASONING HOP LINEAGE
No reasoning failures detected in this run.

Full results → results.jsonl
=====
(base) vanshgala@Vanshs-MacBook-Air physics_stress_tester % █
```

MEDICAL DOMAIN

```
=====
FILE      : results_medical_diagnostics.jsonl
Total steps : 200
=====

— Overall Failure Rates —
Strict hallucination rate : 3.0% 95% CI [1.4% - 6.4%]
Combined failure rate    : 6.0% 95% CI [3.5% - 10.2%]
(Combined = hallucination OR inconsistency across paraphrases)

— Failure Type Breakdown —
RAG hallucination : 0 (0.0%)
Inconsistency     : 7 (3.5%) ← model gives different answers to rephrased Qs
Evasion           : 5 (2.5%)

— Per-Category: Strict Hallucination Rate —
Category      Rate  CI (95%)      n  Verdict
-----
factual       9.2% [ 4.3% - 18.7%]  65  LOW RISK
misleading    0.0% [ 0.0% - 7.9%]  45  RELIABLE
reasoning     0.0% [ 0.0% - 7.9%]  45  RELIABLE
ambiguous     0.0% [ 0.0% - 7.9%]  45  RELIABLE

— Per-Category: Combined Failure Rate (hall. + inconsistency) —
Category      Rate  CI (95%)      n  Verdict
-----
factual       12.3% [ 6.4% - 22.5%]  65  LOW RISK
misleading    2.2% [ 0.4% - 11.6%]  45  RELIABLE
reasoning     2.2% [ 0.4% - 11.6%]  45  RELIABLE
ambiguous     4.4% [ 1.2% - 14.8%]  45  RELIABLE

— Bandit Pull Distribution —
factual       65 pulls (32.5%)
misleading    45 pulls (22.5%)
reasoning     45 pulls (22.5%)
ambiguous     45 pulls (22.5%)
```



CROSS-DOMAIN FINDINGS & INSIGHTS

The evaluation reveals that LLM reliability is highly dependent on the **domain** and **type of reasoning** required.



1 FINANCE DOMAIN

- Hallucination: **Moderate**
- Inconsistency: **High**
- Evasion: **Present**

Weakest Area:
Factual Reasoning

Insight:
Unstable reasoning in real-world financial queries.

Overall Hallucination Rate	Samples
20.0%	30



2 GENERAL / EDUCATION DOMAIN

- Hallucination: **Moderate**
- Inconsistency: **Moderate**
- Evasion: **Low**

Behavior:
Balanced Failure Distribution

Insight:
Mix of hallucination and consistency; stable but not highly reliable.

Overall Hallucination Rate	Samples
6.0%	50



3 MEDICAL DOMAIN

- Hallucination: **Low**
- Inconsistency: **Moderate**
- Evasion: **Present**

Behavior:
Model tends to avoid answering in uncertain scenarios.

Insight:
Safer behavior: evasion instead of hallucination in high-uncertainty cases.

Strict Hallucination Rate	Samples
3.0%	200



4 PHYSICS DOMAIN

- Hallucination: **Very Low**
- Inconsistency: **Low**
- Evasion: **Minimal**

Behavior:
Highly consistent and reliable in structured domain.

Insight:
Structured knowledge leads to stable and reliable outputs.

Overall Hallucination Rate	Samples
4.0%	200



HALLUCINATION PATTERNS

- More frequent in real-world domains (e.g., finance)
- Lower in structured domains (e.g., physics)



INCONSISTENCY BEHAVIOR

- Emerges when multiple responses are generated
- Prominent in domains requiring reasoning or interpretation



EVASION TRENDS

- Appears in high-uncertainty domains (e.g., medical)
- Model avoids answering rather than hallucinating



DOMAIN DEPENDENCE

- Structured domains → more stable outputs
- Real-world domains → higher variability and failure



ADAPTIVE TESTING INSIGHT

- UCB bandit successfully identifies weak areas.
- Focus shifts dynamically toward failure-prone categories.
- Improves efficiency of testing and reliability of evaluation.



KEY INSIGHT:

LLM performance is not uniform — it **varies significantly across domains**, making **reliability evaluation** more important than simple accuracy.



KEY TAKEAWAYS

From Stress Testing Across Domains



1 LLMs are not inherently reliable

- They can hallucinate, contradict themselves, or avoid answering.



2 Reliability is domain-dependent

- Structured domains (e.g., physics) → more stable outputs.
- Real-world domains (e.g., finance) → higher variability.



3 Accuracy alone is insufficient

- Models may appear correct but still behave inconsistently.



4 Multi-response evaluation is critical

- Asking the same question multiple times reveals hidden failures.



5 Multi-hop reasoning exposes deeper weaknesses

- Reasoning-based questions require multiple logical steps.
- Models often fail when chaining concepts across steps.



6 Effect of Sample Size

- As the number of evaluation steps increases, hallucination rates stabilize and often **decrease**.
- Larger sample sizes provide more reliable and statistically consistent estimates.

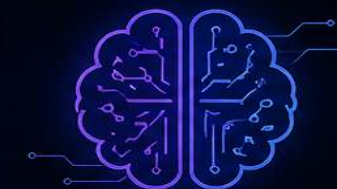


7 Adaptive testing improves evaluation

- UCB bandit efficiently identifies weak areas.
- Focus shifts dynamically toward failure-prone categories.



OVERALL INSIGHT



LLM performance is not uniform — it varies significantly across domains, **reasoning depth**, and **evaluation scale**.



What This Means

Evaluating **trustworthiness** requires more than accuracy — it requires systematic, adaptive, and reasoning-aware stress testing.



FINAL INSIGHT

LLM evaluation must go beyond accuracy and focus on **trustworthiness** under **real-world conditions**.

